Predicting Wildfires in California Using Machine Learning

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Abstract

Wildfire ignition prediction is a critical challenge for fire protection agencies seeking to allocate resources before fires start. This project applies supervised binary classification to predict whether a wildfire will ignite on a given day in California using 14,976 daily weather observations spanning 1984 to 2025. The input to our models is a set of 23 weather and temporal features including temperature, precipitation, wind speed, drought duration, and seasonality indicators. We implement and compare four models: logistic regression, XGBoost, Random Forest, and a feedforward neural network (FFNN), against a majority class baseline. All models substantially outperformed the baseline (66.82% test accuracy, zero fire detection), with XGBoost as our top performing model. Seasonal subgroup analysis revealed that all models perform well in summer driven by high base fire rates but fail to detect winter and spring fires where weather features alone provide insufficient discriminating signal. These findings suggest the predictive ceiling of tabular weather data has been reached, and that future improvements require spatially resolved data, more observations and additional environmental features.

I. Introduction

Wildland fire is a natural part of many ecosystems across the world. Environmental conditions such as temperature, wind speed, and precipitation are critical contributors to how these fires have started. This project explores how modeling real-world wildfire risk can better capture these relationships and classify fire risk on a given day from its weather conditions to support early-warning systems and wildfire risk assessment. The input to our models is daily weather observations for California, and we use logistic regression, XGBoost, Random Forest, and a Feedforward neural network (FFNN) to predict whether a wildfire will ignite on a given day.

II. Related work

Prior studies have used classification and deep learning models on satellite images and weather data to predict wildfires, requiring sophisticated modeling and satellite imagery data that may not be readily available. Other studies focused on fire and smoke to detect when a fire has started. Our study will determine if tabular weather data can be effective at predicting the day a fire will begin, allowing firefighters and authorities to plan for helping to determine fire probability.

One focus area of prior studies has been on predicting forest fire spread [1], [2] using machine learning techniques. For example, Khennou and Akhloufi [1] used convolutional neural networks (CNN) to predict high-risk wildfire areas and found 94.6% accuracy, improving on past literature by 3.7%, including temperature, topography and satellite map data to predict burned areas. The study's strength lies in the use of images, with topographical data adding a third, accurate dimension to their data. The one challenge with their study is that it modeled fire spread and therefore did not predict the fire before it has started.

Radke et al. [2] focused on the growth of the wildfire using supervised models on GIS data (geographic information systems) and tabular data (elevation, weather data, historical fire perimeters). Using a 2D CNN, their model predicted the growth of the wildfire given historical data, location and atmospheric data and reached an accuracy of 87.7%. This study's strength lay in the diversity of data, ranging from location and a larger array of climate data from NOAA. However, a continued limitation of this prior work has been not on the prediction of fire ignition that can inform preventative measures, but rather on examining the spread of the fire post ignition.

The crucial role of weather data, such as temperature, wind, precipitation patterns is important for response agencies to predict how and where fires will spread [3], [4]. However, our study's goal is to focus on weather characteristics that can predict fire ignition prior to wildfires starting, thus increasing the precision and speed of response of fire protection agencies. Decision Tree, Random Forest, Neural Networks and Support Vector Machines (SVM) [5], [6] have been historically used to achieve that goal. Our study extends this to include Feed Forward Neural Networks (FFNN), Gradient Boost with additional weather-related data (e.g., days without rain).

III. Data

A. Dataset

The original dataset contained 14,988 observations spanning 1984 to 2025. The dataset includes four daily weather features (*precipitation*, *max_temp, min_temp*, *avg_wind_speed*), two temporal variables (*month*, *season*), three engineered features (*temp_range*, *wind_temp_ratio, day_of_year*), and two additional features (*lagged_precipitation*, *lagged_avg_wind_speed*). The binary target variable *fire_start_day* indicates whether a wildfire started on a given date. The dataset can be found at http://zenodo.org/records/14712845.

B. Data Preprocessing

We converted the categorical variables of season and month to sets of one-hot encoded features. We engineered an additional feature to capture drought conditions by counting the number of consecutive days without precipitation leading up to each observation. As we treat each day as an independent observation, we removed twelve rows missing data without impact, yielding a final dataset of 14,976 observations.

We used correlation analysis to identify highly correlated features and ensured we were not excluding variables with unique predictive power. Two *temporal* values were excluded: *date and day of year* were removed as temporal effects are captured by *month* and *season.* We also excluded *year* as climate conditions are largely consistent year to year. *Derived* values excluded were *temp_range,* largely redundant as derived from *max_temp* and *min_temp,* and *wind_temp_ratio* as it was derived from *avg_wind_speed* and *temperature variables.*

Our final feature set consists of seven numeric weather features: precipitation, days_without_rain, max_temp, min_temp, avg_wind_speed, lagged_avg_wind_speed, lagged_precipitation, and 16 one-hot encoded temporal features (4 seasons + 12 months) for a total of 23 predictive features.

C. Outlier Detection and Treatment

Detection: using the Interquartile Range (IQR) (Tukey's rule) method with a 1.5 x IQR threshold. *Treatment* - we applied winsorization by capping at the 1st and 99th percentiles, preserving extreme values while reducing their influence.

The discrepancy between outlier detection rates and winsorization adjustment rates occurs because the IQR method is more conservative than percentile capping used in winsorization, which only adjusts the most extreme 2% of values. This approach is important for weather data, where extreme values (e.g., heat waves, heavy rainfall) may represent genuine conditions associated with elevated fire risk. See appendix.

D. Feature Scaling

During preprocessing, feature scaling was not applied as they will be applied during model training to avoid unnecessarily transforming data for algorithms that don't benefit from it. Logistic Regression and Feedforward Neural Network will use standardization (Z-score normalization) to ensure gradient descent converges efficiently. Gradient Boost and Random Forest will not use scaling, as tree-based models are invariant to monotonic transformations of features.

IV. Exploratory Data Analysis (EDA)

A. Distribution Analysis and Multicollinearity Assessment

We examined pairwise scatter plots and calculated Pearson correlation coefficients between features. *Temperature* variables (*max_temp, min_temp*) show strong positive correlation (r = 0.64), as expected. *Precipitation* (*precipitation* and *lagged_precipitation*) show moderate autocorrelation between current and lagged values (r = 0.47), suggesting that 7-day rainfall patterns persist. *Wind speed* shows strong autocorrelation (r = 0.60) between avg_wind_speed and lagged_avg_wind_speed, indicating wind patterns are relatively stable week-to-week.

B. Correlation with Target Variable

We analysed Pearson correlation coefficients between all numeric features and fire_start_day, revealing three features with strong positive correlations.

First, *min_temp* shows the strongest correlation (r = 0.47), indicating that warm overnight temperatures are the single best linear predictor of fire ignition. Second, *max_temp* shows positive correlation (0.37), confirming that daytime heat contributes significantly to fire risk. Last, *days_without_rain* shows positive correlation (0.31), demonstrating that prolonged drought periods increase fire susceptibility. Additionally, moisture-related features show predictive effects with moderate negative correlations. *Lagged_precipitation* (7-day prior rainfall) shows moderate negative correlation (-0.21), *Precipitation* (same-day rainfall) shows weaker negative correlation (-0.12).

Temporal features show modest to low correlations, with month exhibiting low positive correlation (r = 0.14). Wind speed variables (*avg_wind_speed*, *lagged_avg_wind_speed*) show limited relationship (r = 0.03-0.08), suggesting that while wind can contribute to fire spread, it is less important for initial ignition compared to temperature and moisture conditions.

C. Data Visualizations

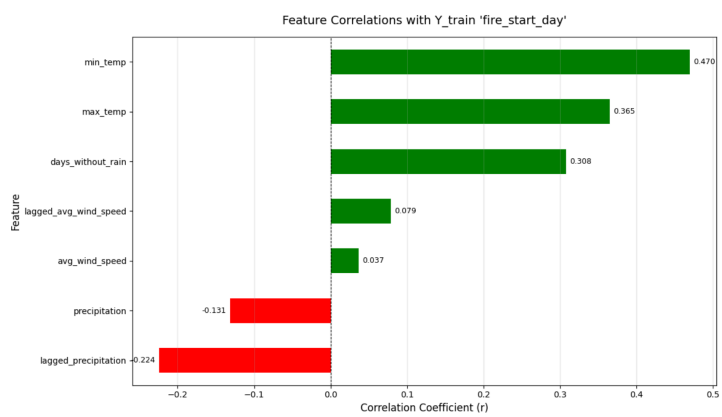

Fig. 1. Correlations of Weather Features and Fire Risk

Fig. 1. highlights the correlation strength between each individual feature and *fire_start_day*, demonstrating that temperature variables and drought conditions exhibit the strongest predictive relationships with fire occurrence, while moisture-related features show protective effects.

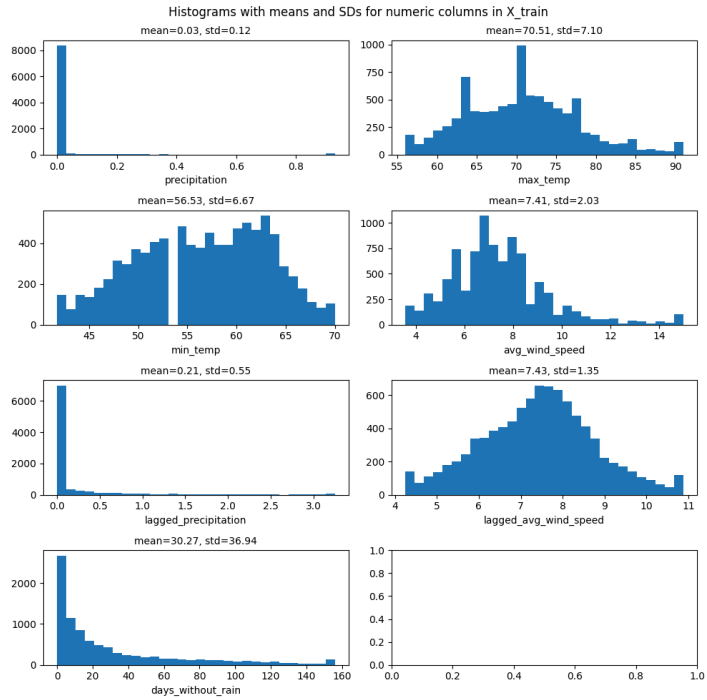

Fig. 2. Distribution of Weather Features in California Fire Dataset (1984-2025).

Fig. 2. histograms demonstrate the distribution of seven numeric weather features and two temporal features across 14,976 daily observations. The right-skewed distributions of precipitation, lagged_precipitation, and days_without_rain, reflect California's Mediterranean climate with extended dry periods punctuated by concentrated rainfall events. This skewness motivated our use of winsorization rather than removal for outlier treatment, as extreme precipitation events represent genuine weather phenomena rather than measurement errors.

D. Challenges

A noted limitation of the dataset is that observations do not contain geographic identifiers (e.g., latitude, longitude, weather station ID/location, or regional designation), preventing analysis of spatial variation in fire risk across the diverse state climate zones. California encompasses dramatically different environments, each with distinct fire risk profiles and weather patterns. Thus, our models cannot account for location-specific fire dynamics. Our predictions instead represent statewide average relationships between weather and fire ignition, potentially obscuring important regional differences.

V. Methodology

Our goal is to predict whether a wildfire starts on a given day. Each day is labeled as a fire day (1) or non-fire day (0), making this a supervised binary classification problem. Our baseline is a majority class classifier that predicts the majority class - no fire - for each observation, regardless of all other conditions. With 67% of days in the dataset being non-fire days, this simple approach achieves 67% accuracy without learning any patterns from the data. While the baseline establishes a performance floor that all subsequent models must exceed to demonstrate value, it is operationally useless - it catches zero fire days and offers no predictive signal.

We split the data into training (60%, 8,985 observations), validation (20%, 2,995 observations), and test (20%, 2,996 observations) sets using stratified sampling. We trained each model, handled class imbalance, tuned parameters on the validation set, and compared their performance on the test set. All models were evaluated on accuracy, precision, recall, F1-score, and area under the precision-recall curve. The final model was selected based on the highest F1-score, with recall as the tiebreaker.

Logistic regression learns a weight for each of the 23 input features and combines them linearly, then applies a sigmoid function to produce a probability between 0 and 1. If that probability exceeds 0.5 the day is classified as a fire day. Because it assumes a linear decision boundary, logistic regression is well suited as a first improvement over baseline - it is interpretable, computationally efficient, and provides a strong reference point for evaluating whether more complex models add value. Standardized features were used since gradient descent converges more efficiently when inputs are on a common scale. Class weighting was applied in the TensorFlow training loop to address the 67/33 class imbalance.

XGBoost is a gradient boosting algorithm that builds sequential decision trees, where each new tree corrects the residual errors of the previous one. This sequential error correction allows XGBoost to capture complex nonlinear relationships and feature interactions. Raw unstandardized features were used as tree-based models are invariant to feature scale. One-hot encoded season and month columns were replaced with a single integer month feature, reducing the feature set from 23 to 8. Class imbalance was addressed using scale_pos_weight of 2.01, reflecting the 67/33 no-fire/fire ratio. Hyperparameters were tuned using RandomizedSearchCV over 1,000 candidate configurations and 3-fold cross-validation, searching across 12 parameters including max_depth, learning_rate, n_estimators, regularization terms and subsampling rates. The optimal number of boosting rounds was determined by training to 10,000 rounds and identifying the point of minimum validation loss, which occurred at 6,938.

The **Feedforward Neural Network** (FFNN) passes inputs through multiple layers of learned weights and nonlinear activation functions, enabling it to learn complex patterns not captured by linear or tree-based models. With built-in TensorFlow/Keras tools, the design consists of an input layer accepting all 23 standardized features, between 0 and 3 hidden dense layers, and a final sigmoid output layer producing a fire probability between 0 and 1. Z-score normalization was applied to numeric features using training set statistics to prevent data

leakage. Class imbalance was addressed by applying class weights of 1 and 2 for no-fire and fire days respectively during training. The model was trained for 100 epochs with a batch size of 32 using binary cross-entropy loss, with a threshold of 0.5 applied to probabilities to generate final binary predictions.

Random Forest is an ensemble method that builds multiple independent decision trees on bootstrapped subsets of the training data and aggregates their predictions by majority vote. Each tree is trained on a different random sample of observations and a random subset of features at each split, introducing diversity that reduces overfitting relative to a single decision tree. Raw unstandardized features were used since tree-based models are invariant to feature scale. *class_weight* = 'balanced' was applied to up-weight the minority fire class during tree construction. Hyperparameters were tuned by manually iterating over candidate values of *n_estimators* and *max_depth* - the two key RF parameters - and selecting the combination with the highest F1-score on the validation set.

VI. Experiments and Results

The **logistic regression** model hyperparameters were tuned using Keras Tuner Hyperband, searching over learning rates (0.01, 0.001, 0.0001, 0.00001) and batch sizes (16, 32, 64). The optimal configuration was learning_rate = 0.01 and batch_size = 64, with the best epoch determined at epoch 43. This achieved a test accuracy of 76% which is 8% better than the baseline guessing of the majority class, no fire, every time, at 67%. The model caught 79% of all fires (fire recall). The fire precision (60%) means that 40% are false alarms.

XGBoost. After tuning, the best cross-validated F1 was 0.684. The optimal number of boosting rounds was then identified by training to 10,000 rounds and selecting the point of minimum validation loss at round 6,938. The final model achieved test accuracy of 76.47%, fire recall of 78.17%, fire precision of 61.42%, fire F1 of 68.79%, and PR-AUC of 0.69. Training loss (0.420) was lower than validation loss (0.497), with a gap of 0.077, confirming moderate overfitting that was mitigated through L1 and L2 regularization, row and column subsampling, and minimum child weight constraints.

Feedforward Neural Network. The FFNN architecture was tuned using Keras Tuner, simultaneously searching over the hidden layers (0-3), units per layer {16, 32, 64, 128}, activation function {ReLU, tanh}, and learning rate {0.01, 0.001, 0.0001}. The model with optimized hyperparameters achieved test accuracy of 74.50% fire recall of 78%, fire precision of 59%, and fire F1 of 67%. Training and test loss were similar (0.48 vs 0.53), suggesting no significant overfitting. The model's fire recall was 78%, comparable to other models, and the fire precision is 59%, also comparable to other models.

Random Forest. The tuned Random Forest (n_estimators = 100, max_depth = 5) achieved 75.60% test accuracy, fire recall of 77.00%, fire precision of 60%, and fire F1 of 0.68. We see an F1 score on the no-fire set of 80% (vs. 68% on the fire set), so the model's performance (measured by F1) on the majority class is higher, as expected. Hyperparameter tuning revealed a clear overfitting pattern. Deeper trees consistently degraded validation F1. Overall, the model is reasonably good at catching real fires (77% recall) but raises too many false alarms (40% of its fire predictions are wrong). Overall accuracy of 75.60% represents a meaningful improvement over the majority class baseline of 66.82% (always predicting no fire).

Overfitting Analysis. All four models showed some degree of train-test gap, with Random Forest showing the largest difference in raw accuracy (79.92% train vs 76.60% test) due to tree memorization. XGBoost showed moderate overfitting in loss (gap of 0.077) mitigated by regularization. Logistic regression generalized most cleanly with the smallest train-test loss gap (0.013). The FFNN showed negligible overfitting in loss.

Model Comparison and Selection. Under the project's model selection criteria of highest F1-score, XGBoost is barely outperforming the other models with a fire F1 of 68.79%, marginally edging logistic regression (68.03%) and Random Forest (68.00%). All models substantially outperform the majority class baseline of 66.82% accuracy, which catches zero fire days by always predicting no fire. XGBoost's subgroup analysis (see Fig. A2 in Appendix) shows that it performs better in summer months compared to the winter months. The model missed nearly all winter fires, where events are rare and the model defaults to predicting no fire. It had the highest test accuracy across seasons. The model generalizes well with a minimal train-validation loss gap (0.077), confirming low overfitting. We see a similar precision-recall for all models (see Fig. A1 in Appendix).

XGBoost, a tree-based model, likely outperformed the other models by efficiently capturing non-linear interactions (like seasonal splits) that Logistic Regression missed as it is a linear model. By pruning branches with insufficient gain, XGBoost is often more surgical than Random Forest, which typically grows trees to a fixed max_depth. Additionally, while the FFNN model is often better suited for video, audio, and image data and typically benefits from a larger feature set, the other three models can handle one-hot encoded and tabular data better. Finally, XGBoost focused on reducing overall loss, navigating the class imbalance (winter vs. summer fires) by correctly identifying "no fire" days, as the most efficient way to minimize overall loss.

Table: Evaluation Metrics for All Five Models

Metric	Baseline	Log. Reg.	Rand. Forest	FFNN	XGBoost
Training Accuracy	66.80%	75.59%	76.19%	76.71%	79.92%
Test Accuracy	66.82%	75.47%	75.60%	74.50%	76.47%
Training Loss	0.8705	0.5035	NA	0.4842	0.42
Test Loss	0.8707	0.5035	NA	0.5285	0.4899
Precision (F)	0.00%	59.92%	60.00%	59.00%	61.42%
Recall (F)	0.00%	78.67%	77.00%	78.00%	78.17%
F1 (NF)	80.11%	80.09%	80.00%	79.00%	81.00%
F1 (F)	0.00%	68.03%	68.00%	67.00%	68.79%

Subgroup analysis was conducted for Random Forest and XGBoost to evaluate model performance across seasonal and monthly segments respectively. Seasonal analysis revealed that summer performance (recall 99.05%, F1 81.6%) is driven primarily by the 69% base fire rate rather than genuine discrimination, fall was more balanced (recall 70.9%, F1 55.2%), spring was operationally concerning with recall of only 35.8%, and winter showed complete failure with zero fires detected despite class balancing. For XGBoost, monthly subgroup analysis showed similarly strong performance in summer months where fire base rates are highest, but near-complete failure on winter fires where low event frequency caused the model defaulting to predicting no fire. The consistent finding is that models have learned broad seasonal patterns but lack the ability to discriminate fire risk within seasons, particularly in winter and spring.

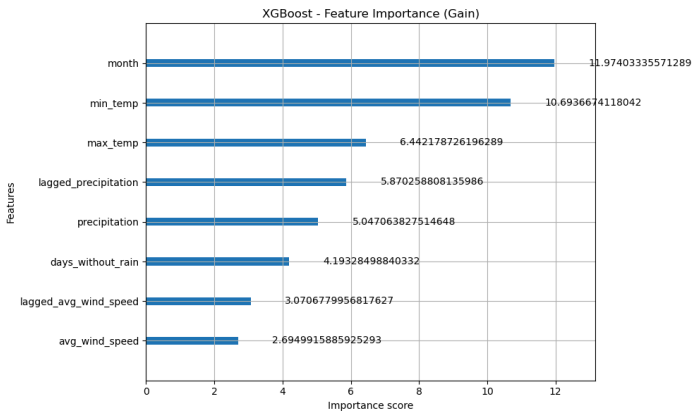

Fig. 3. Feature Importance from XGBoost model

VII. Discussion

Our study aimed to predict fire ignition prior to wildfires starting by focusing on weather data, such as temperature, wind, precipitation patterns. The study compared the findings across Logistic Regression, Random Forest, FFNN, and XGBoost. All four models improved substantially over the majority class baseline, demonstrating that daily weather features carry genuine predictive signals for wildfire ignition. However, the convergence of performance across models - with logistic regression, XGBoost, and Random Forest all landing near 76% accuracy and fire F1 of 68-69%, suggests the primary limiting factor is the data itself rather than model complexity.

Feature importance for all models highlighted that the most gain was made by *min_temp, month*, *max_temp* and *lagged_precipitation*. In other words, night-time temperature, dryness, heat and summer months were the dominant predictors (see Fig. 3). High nighttime temperatures during fire-prone months serve as a more robust predictor of ignition than peak afternoon temperatures. This suggests that the lack of overnight cooling is a critical driver of fire risk, whereas cooler nights effectively lower the probability of ignition the subsequent day. This finding is consistent with emerging literature that trees are unable to recuperate without cooler temperatures at night, especially during summer months, and are more susceptible to catching forest fires or ignition [7].

The seasonal subgroup analyses reinforce this interpretation: all models perform well in summer where fire base rates are high, but struggle in winter and spring where genuine discrimination from weather features alone appears insufficient. Future work incorporating spatial features, fuel moisture data, or ignition source information could meaningfully improve performance in these underperforming seasons.

VIII. Conclusion

This project[1] demonstrated that daily weather data can classify California wildfire ignition days with meaningful accuracy. XGBoost was the best performing model under the project's F1 and recall criteria, achieving 76.47% test accuracy and 78.17% fire recall. The relative consistency of results across logistic regression, XGBoost, and Random Forest suggests that the signal in the available features has been largely captured, and that future performance gains require richer data rather than more complex models. Key limitations include the inability to account for regional climate variation, the lack of features capturing climate-level variation for data points across years, and the models' shared failure to detect winter fires. With access to location-specific weather stations, vegetation dryness indices, and historical ignition data, a more granular model could better serve early warning use cases. A higher volume of data would also increase generalizability.

IX. Contributions

Alvaro Laham: GitHub repo, data preprocessing, feature engineering, methodology; baseline and XGBoost; Maintained: *eda_and_preprocessing_xgboost .ipynb* (Alvaro branch).

Ruchi T. Bhanot: Supported EDA, data shuffling and splitting, and logistic regression code. Related work, Methodology (Logistic Regression), Results and Discussion. Maintained: *eda_and_preprocessing_Ruchi.ipynb* (ruchit branch)

Jake Klein: Supported EDA/preprocessing code, the outlier analysis, as well as building and evaluating the Feed Forward Neural Network. Maintained: *eda_and_preprocessing.ipynb* (main branch), and *Model2_ffnn.py, Model2_ffnn_best_model_eval.py* (ffnn branch)

Phillip Dundas: Abstract, outlier analysis, correlation analysis, data, methodology and experiment; Random Forest model, seasonal subgroup analysis; maintained *eda_and_preprocessing_Phil_Random_Forest.ipynb* (phild73 branch)

[1] GitHub repo: https://github.com/alvarolaham-ucb/207_final_project_Alvaro_Jake_Ruchi_Phillip

References

[1] Khennou, Fadoua, and Moulay A. Akhloufi. "Improving wildland fire spread prediction using deep U-Nets." *Science of Remote Sensing* 8 (2023): 100101.

[2] Radke, David, et al. "FireCast: Leveraging deep learning to predict wildfire spread." *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence (IJCAI-19)* (2019): 4575-4581.

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[4] Agarwal, Puneet, et al. "Big data and predictive analytics in fire risk using weather data." *Risk Analysis* 40.7 (2020): 1438-1449.

[5] Sakr, George E., et al. "Artificial intelligence for forest fire prediction." *IEEE/ASME International Conference on Advanced Intelligent Mechatronics* (2010): 1311-1316.

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APPENDIX

Visualizations

Table A1: Our outlier analysis revealed the information as follows:

FEATURE	OUTLIERS DETECTED (TUKEY'S RULE)	VALUES ADJUSTED POST-WINSORIZATION
PRECIPITATION	9.22%	0.99%
MAX_TEMP	1.08%	1.58%
MIN_TEMP	0.01%	1.78%
AVG_WIND_SPEED	3.20%	1.75%
LAGGED_PRECIPITATION	20.47%	0.99%
LAGGED_AVG_WIND_SPEED	1.23%	2.00%
DAYS_WITHOUT_RAIN	7.39%	0.96%

Fig. A1: Precision-Recall curves for each evaluated model. We see comparable performance for all models, tracing similar curves with comparable overall average precisions

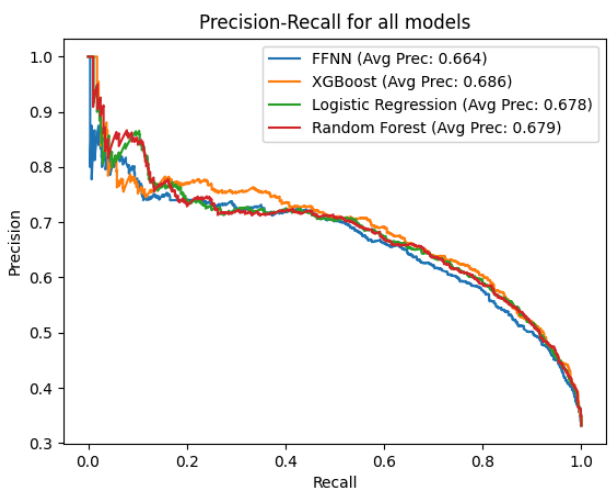

Fig. A2: XGBoost By-Month Performance Analysis

We conduct an analysis of performance by month for the XGBoost model. Full results can be found here, and a summary of results are shown in the plot below. We see a higher recall in summer months, with lower recall (or 0 recall) in winter.

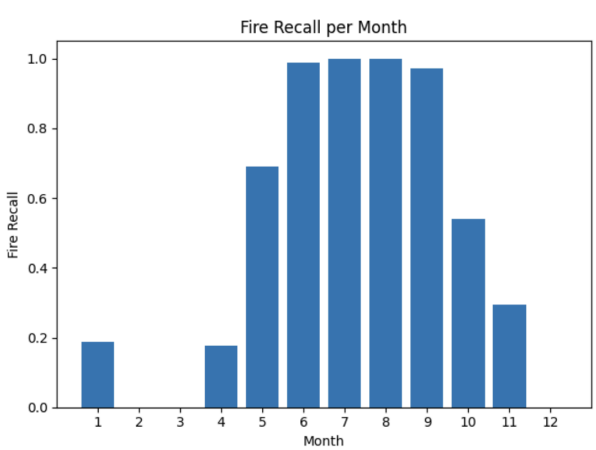

Fig. A3: Training process for XGBoost model

Below shows an example of the training procedure for one of our models, the XGBoost model.

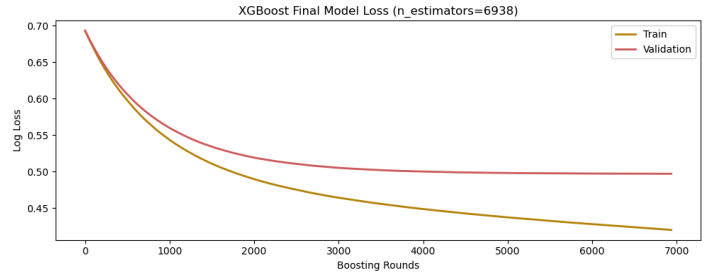
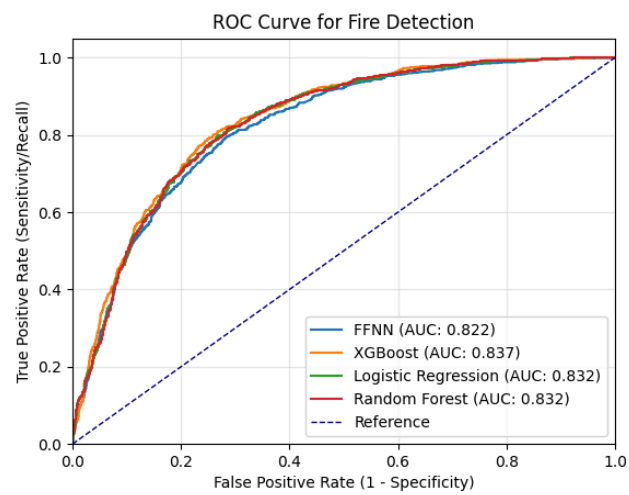
Fig. A4: Receiver Operating Characteristic (ROC) Curve for all models evaluated